

Cluster Training: A Novel Method for Introducing Training Program Variation

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SUMMARY

THE INTRODUCTION OF NOVEL TRAINING STIMULI PLAYS A CRUCIAL ROLE IN INDUCING SPECIFIC TRAINING ADAPTATIONS. ONE METHOD THAT CAN BE EMPLOYED TO INTRODUCE A NOVEL STIMULUS TO THE TRAINING PROGRAM WHILE MAXIMIZING THE VELOCITY AND POWER OUTPUT OF THE TRAINING EXERCISE IS THE INCLUSION OF THE CLUSTER SET CONFIGURATION. THE CURRENT REVIEW PRESENTS THE THEORETICAL AND RESEARCH FOUNDATION FOR THE USE OF THE CLUSTER SET IN PERIODIZED TRAINING PROGRAMS AND OFFERS EXAMPLES OF PRACTICAL APPLICATIONS THAT CAN BE USED IN THE PREPARATION OF ATHLETES IN A VARIETY OF SPORTS.

INTRODUCTION

One of the key concepts of periodization is that programs are designed to introduce appropriate training variation in a logical and systematic fashion in an attempt to stimulate improvements in some performance or physiological outcome.

Training variations are essential because they stimulate recovery and adaptation, the avoidance of overtraining, long-term phase potentiation, and an elevation in performance outcomes (17). Variation can be introduced into a periodized training program in many ways. Some typical examples of training variations that can be employed when designing a periodized program are manipulations of the overall training load, number of sets, number of repetitions, set configurations, and the exercises selected. These potential methods for introducing training variation allow the strength and conditioning professional a means for introducing novel stimuli into the training program. Hodges et al. (10) suggest that the introduction of novel stimuli allows a more rapid gain in performance and that the more familiar the individual is with the task, the slower the overall gains in performance are. Therefore, it is essential that the strength and conditioning professional employs variations in the overall training program design in order to maximize the training outcomes. This is especially true for advanced and elite athletes.

One often overlooked method of employing variation to the training program is the manipulation of the

structure of the set being employed. Traditionally, the configuration of a set requires the athlete to perform each repetition in a continuous fashion where no rest is taken in between each repetition of the set (5,9,22). Recently, an additional type of set configuration termed the rest-pause set (5) or cluster set (9,21) has been proposed as a way of altering the structure of a training set. In this type of set configuration, an interrepetition rest interval of 10–30 seconds is employed between each repetition performed (9). The configuration of the cluster set can be manipulated in several ways that may include using variable rest interval durations or manipulating the resistance used with each repetition of the cluster set depending on the purpose or the focus of the current block of training employed in the periodized training program. There are generally two types of intensity modification that can be employed with cluster sets, the undulating and the ascending cluster set configuration. In the undulating cluster set, the resistance is increased in a pyramid type fashion (9), while during

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the ascending set configuration, the resistance is increased with each successive repetition. When formulating the different methods of manipulating set configurations, each type of set configuration should be considered in regard to the overall training plan. Additionally, the strength and conditioning professional should consider the overall goal of each phase of training when attempting to employ various set configurations.

The purpose of this brief review is to discuss the theoretical basis for the use of the cluster set configuration, present scientific evidence that examines the use of the cluster set, and give practical examples of how a cluster set might be employed in a periodized training program.

THEORETICAL BASIS FOR CLUSTER SETS

The use of short rest intervals between the individual repetitions of a set should theoretically result in improved quality of performance during each of the repetitions (9). In 2003, Haff et al. (9)

presented a hypothetical model for the effects of cluster sets on performance. In this model, it was suggested that performance characteristics such as peak power output, barbell velocity, and displacement would decrease with each subsequent repetition of a traditional set where no interrepetition rest was used (Fig. 1). The concept of an interrepetition rest interval or cluster was suggested as a method for allowing each repetition of the set to be performed with the highest quality. Therefore, it was hypothesized that the inclusion of a cluster set configuration in which 15–30 seconds of recovery would be employed between repetitions would allow the individual to experience partial recovery and thus perform each repetition with a higher power output, peak barbell velocity, and peak barbell displacement.

When considering the potential of the cluster set configuration for increasing the individual repetition power, it is possible that an increase in the average power output (Fig. 2) of a training set occurs (14). The use of a cluster set

paradigm may be beneficial in the development of power-generating capacity as it may result in a decrease in repetition-induced fatigue (14,18). When a set is performed in the traditional fashion, it is likely that interrepetition fatigue may manifest itself as acute fatigue factors associated within the neuromuscular system or by the accumulation of metabolic fatigue inducing factors, ultimately resulting in a decrease in repetition power.

Viitasalo and Komi (24) have reported that reductions in maximal force-generating capacity, rate of force development, and rate of relaxation can occur in as few as 5 to 9 maximal contractions. They hypothesized that increases in blood lactate were partially responsible for the fatigue-induced alterations in maximal force-generating capacity and selected force-time curve characteristics. Hypothetically, the inclusion of a 15- to 30-second interrepetition rest interval may result in some phosphocreatine (PCr) replenishment, while traditional sets configurations result in greater PCr depletion, which

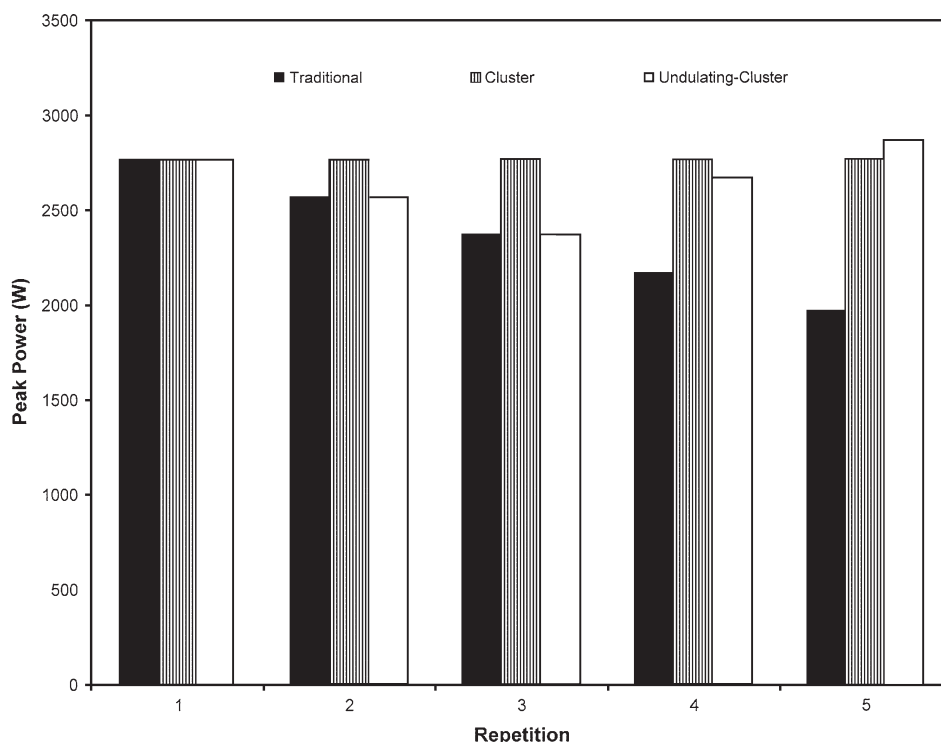


Figure 1. Hypothetical model of peak power responses to traditional, cluster, and undulating cluster set configurations.

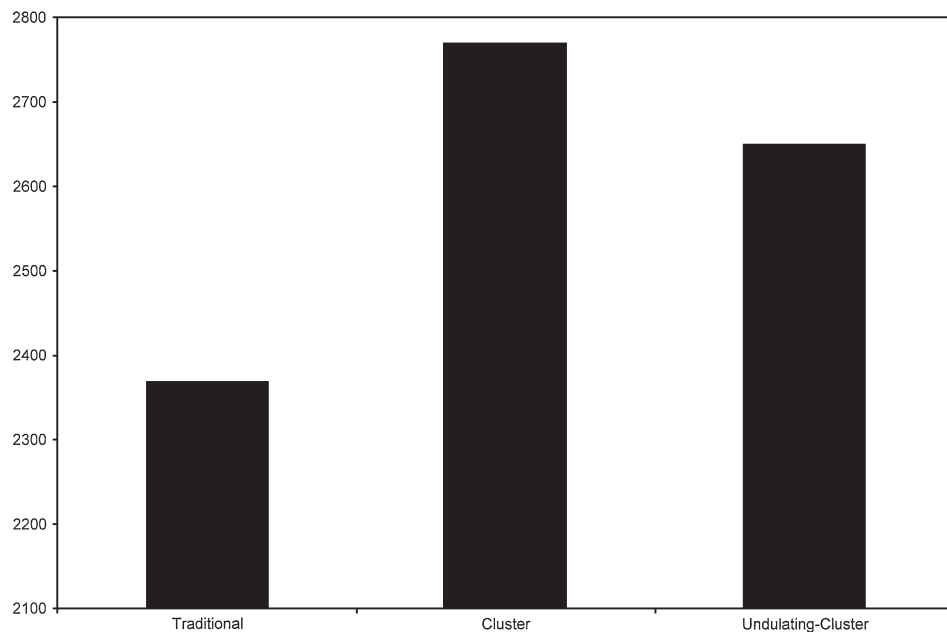


Figure 2. Hypothetical model of average peak power during a traditional, cluster, and undulating cluster set of 5 repetitions.

ultimately stimulates an increased production of lactic acid and lactate as the athlete uses more muscle glycogen (9). Some support for this contention can be gained from the work of Sahlin and Ren (19) who reported that maximal contractions resulted in significant decreases in both adenosine triphosphate (ATP) and PCr concentrations. Decreases in both ATP and PCr were associated with significant elevations in lactate concentrations, which corresponded to substantial decreases in the amount of force that can be generated. The addition of 15 seconds of recovery resulted in an increase in maximal force-generating capacity that corresponded to $\sim 79.7 \pm 2.3\%$ of initial capacity (19). Similarly, when 30 seconds of extra recovery was used between 5 maximal cycle ergometer sprints, a significant elevation of peak power-generating capacity and a reduction in lactate formation were noted (26). The elevations in lactate associated with the shortest rest intervals are generally associated with negative effects on muscle contraction as a result of impairments in ATP generation that result in changes in contractile characteristics, which ultimately alter performance

outcomes (19). Based on this line of reasoning, the use of cluster sets might be a superior method for enhancing muscular strength, power, and growth.

While the conceptual model of employing a cluster set configuration appears to be a sound model for developing maximal strength, enhancing power-generating capacity, or stimulating greater hypertrophy, Lawton et al. (15) suggest that the inclusion of a cluster set-loading paradigm may be most beneficial for explosive or ballistic strength training methods such as those used in programs that rely on weightlifting movements. Support for this idea can be found in the work of Rooney et al. (18). Although not all studies agree (6), Rooney et al. (18) suggested that interrepetition rest intervals decrease repetition fatigue, but do not promote the same level of strength gains when compared to traditional set configurations. Additionally, it was suggested that traditional continuous repetition paradigms increase strength development via an increased activation of higher threshold motor units and production of metabolic fatigue-induced muscular adaptations (15,18). Additionally,

Kraemer et al. (13) suggest that lactate production favors a hypertrophic response. Based on this line of reasoning, the cluster set configuration may be most useful for the development of explosive power and more traditional set configurations may be better suited for the development of maximal strength or stimulating hypertrophic responses.

When using a cluster set configuration in an attempt to improve power-generating capacity, structuring the set in an undulating fashion may be one method that has the potential to magnify the adaptive stimulus. The undulating set configuration uses a series of repetitions performed in a cluster format in which the resistance ascends followed by a series of lighter efforts (9). For example, in an undulating cluster set of 5 repetitions, the athlete may perform 3 ascending repetitions (i.e., 85%, 90%, 95% of 1RM) followed by 2 descending repetitions performed with lighter intensities (i.e., 90% and 85%). Theoretically, the descending portion of the undulating cluster should result in a potentiation effect in which greater power outputs, barbell velocities, and displacements are achieved (Fig. 1).

These potential effects may occur as a result of a postactivation potentiation effect. Postactivation potentiation is an enhancement of force seen after repetitive skeletal muscle activation (1). The mechanism behind post activation potentiation, while not completely understood, appears to be the result of an increased phosphorylation of myosin regulatory light chains (20) or a neural effect in intact muscle (3). Increased phosphorylation sensitizes the actin-myosin interaction to Ca^{2+} , which leads to greater force production in skeletal muscle. Neural effects could include increased motor unit synchronization, desensitization of alpha motor neuron input, and decreased reciprocal inhibition to antagonists. While the undulating cluster set may induce a potentiation effect, it should be considered an advanced set modification and may be best suited for highly trained individuals. Based on current literature, postactivation potentiation complexes are most effective when used by well-trained individuals (3). Therefore, it appears that undulating cluster sets may have a greater potential for inducing specific training adaptations when implemented with highly trained individuals.

Collectively, it appears that, from a theoretical standpoint, the inclusion of cluster set configurations has the potential to alter the training stimulus and ultimately magnify the adaptive response. By altering the set configuration, the strength and conditioning professional may have the ability to develop specific adaptive responses that may favor maximal strength, explosive strength and power, or muscular growth.

RESEARCH SUPPORT FOR CLUSTER SETS

To the authors' knowledge, there are only a very few papers that have been published in the peer-reviewed literature that examine cluster sets in either short- or long-term training situations (2,4,9,14,15,18). Summaries of the research that have been concerned with the short- and long-term effects of cluster sets are presented in Tables 1 and 2, respectively.

ACUTE RESPONSES TO CLUSTER SET TRAINING

In 2003, Haff et al. (9) examined the effect of 3 different types of set configurations consisting of a traditional set, cluster set, and an undulating cluster set. The traditional set and cluster set were performed with 5 repetitions at an intensity of 90% and 120% of 1RM power clean. The undulating sets consisted of 5 repetitions performed at an average intensity of 90% or 120% of the subject 1RM power clean. The athlete performed a repetition at 85%, 90%, 100%, 90%, and 85% of the subject's 1RM power clean for an average 5-repetition intensity of 90% or a repetition 110%, 120%, 140%, 120%, and 110% of the subject's 1RM power clean for a 5-repetition average intensity of 120%. The interrepetition rest interval for each of the cluster sets was set at 30 seconds. The 3 different set configurations were tested using the clean pull with 2 intensities of 90% and 120% of a 1RM power clean. When examining the 90% intensity trial, the cluster set exhibited a statistically significant increase in average barbell velocities (+8.1%) and a nonstatistically significant increase in barbell displacement (+5.9%) when compared to the traditional set. While the average peak power output for the set was not significantly different, the cluster set resulted in a 6.8% increase in peak power when compared to the traditional set. For the 120% of 1RM intensity, a statistically significant increase in average peak barbell velocity (+7.9%) and displacement (+2.1%) was noted during the cluster set when compared to the traditional set. Conversely, the average peak power for both the cluster sets was not different (-0.4%) than the traditional set configuration. One rationale for the lack of difference in power output during the 120% intensity cluster sets may be because previously it was reported that a 90% intensity is the optimal load for pulling exercises (8,16), thus potentially confounding the power data. Based on this study, it can be concluded that the use of a cluster set may result in enhancements in velocity of movement,

displacement of the barbell, and, most likely, power-generating capacity.

In order to investigate the effects of set configurations on the acute repetition power outputs during the bench press, Lawton et al. (15) used 4 different set configurations. The 4 sets configurations included (a) a traditional set of 6 repetitions performed at a 6RM intensity with no rest between each repetition, (b) a cluster of 6 singles performed at a 6RM intensity with 20 seconds between each repetition, (c) a cluster of 6 repetitions performed as 3 pairs of doubles with a 6RM intensity and 50-second rest between each pair of doubles, and (d) a cluster of 6 repetitions performed as 2 clusters of triples with a 6RM intensity and 100-second rest between each group of triples. The first major finding of this project was that the traditional set resulted in a linear decrease in power output across the repetition range. These findings support the hypothetical model previously proposed by Haff et al. (9) (Fig. 1). Similar to the results presented by Haff et al. (9), the cluster sets resulted in statistically significant greater individual repetition power output and total power output when compared to the traditional set configuration. However, there were no significant differences between the 3 cluster set configurations with regard to individual repetition power or total power output. Lawton et al. (14) concluded that the cluster set paradigm may be very beneficial for explosive or ballistic strength exercise. Therefore, this type of set configuration may be useful to the strength and conditioning professional who is using weightlifting exercises such as the power clean, power snatch, or pulling motions.

In another investigation, Denton and Cronin (4) examined the kinematic (displacement, velocity, and acceleration), kinetic (force impulse, work, and power), and lactate responses to different set configurations. Three loading schemes were employed in conjunction with the bench press exercise in this investigation for a targeted total of 24 repetitions. The first loading scheme

Table 1
Acute effects of cluster sets

Author	Subjects	Set configurations	Interrep rest interval (s)	Repetition sequence	Intensity	Exercise	Results
Haff et al. (9)	8 male track athletes	Traditional	0	1 set 5	90/120%*	Clean pull	Peak velocity
	5 male weightlifters	Cluster	30	1 set 5/1	90/120%*	Clean pull	C > T
		Undulating cluster	30	1 set 5/1	90/120%*	Clean pull	
Lawton et al. (15)	12 male basketball players	Traditional (T)	0	1 set 6	6RM	Bench press	Peak power
	14 male soccer players	Cluster (single) (C1)	20	1 set 6/1	6RM	Bench press	C1 = C2 = C3
		Cluster (doubles) (C2)	50	1 set 6/2	6RM	Bench press	> T
		Cluster (triples) (C3)	100	1 set 6/3	6RM	Bench press	
Denton and Cronin (4)	9 resistance trained males	Traditional (control) (T)	0	4 sets 6	6RM	Bench press	Repetitions
		Cluster 1 (C1)	130	1 set 24/3	6RM	Bench press	C2 > C1 = T
		Cluster 2 (every other cluster to failure) (C2)	130	1 set 24/3	6RM	Bench press	Mean power
							C2 > C1 = T
							Work C2 > C1 = T
							Lactate
							T > C1
							C2 > T & C1

*The percentages used in testing were based on the subjects 1RM in the power clean.

5/1 = a set of 5 singles was performed for a total of 5 repetitions; 6/1 = a set of 6 singles was performed for a total of 6 repetitions; 6/2 = a set of 3 doubles was performed for a total of 6 repetitions; 6/3 = a set of 2 triples was performed for a total of 6 repetitions; 24/3 = a set of 8 triples were performed for a total of 24 repetitions.

used was composed of 4 sets of 6 RM with a 302-second recovery between groupings (traditional = T). The second set configuration comprised 8 sets of 3 performed with a 6RM load with each group of 3 separated by 130 seconds (cluster 1 = C1). The final grouping was identical to the second with the exception that every other set was performed to volitional failure (C2). The results of the study revealed that the C2 configuration resulted in

significantly greater repetitions (~30) when compared to C1 (~24) and the traditional (~23.6) set configurations. When examining the mean power output, total work and impulse of the sets configurations C2 were significantly higher than both C1 and T, which were not different. The blood lactate response for C2 was consistently higher than both T and C1. The results of this investigation suggest that increasing the interrepetition rest resulted in

the ability to perform more work at a higher quality.

When examining the acute studies, several key conclusions can be drawn from the literature. First, it appears that cluster set training does allow for acute alteration in the overall training stimulus induced by a specific exercise. While more work is needed in this area, it appears that these set configurations are best suited for ballistic

Table 2
Chronic effects of cluster set training

Author	Subjects	Set type	Repetition sequence (sets × reps)	Interrep rest interval (s)	Weeks of training	Intensity	Results
Byrd et al. (2)	50 untrained males	Control (C)	—	—	10 wk, 3 d/wk	6–10RM	↑ Physical work capacity
		Traditional (T)	3 × 6–10	0			CL2 = CL1 > T = C
		Cluster 1 (CL1)	3 × 6–10	1			↑ 1RM bench press
		Cluster 2 (CL2)	3 × 6–10	2			CL2 = CL1 = T > C
							↑ 1RM leg press
							T > CL1
							CL 2 > C
Rooney et al. (18)	18 males	Control (C)	—	—	6 wk, 3 d/wk		↑ Isometric elbow flexion strength
	24 females untrained	Traditional (T)	10 × 6	0		6RM	T = CL > C
		Cluster (CL)	6 × 6	0			↑ Dynamic elbow flexion strength
			10 × 6/1	30			
			6 × 6/1	30			T > CL > C
Lawton et al. (14)	12 male basketball players	Traditional (T)	4 × 6	0	6 wk, 3 d/wk	6RM	↑ Bench press power
	14 male soccer players	Cluster (C)	8 × 3	113			T > C
							↑ Bench press 6RM
							T > C

power exercises such as those used in weightlifting or exercises such as jump squats. Finally, it appears that the cluster set configuration has the potential to increase work capacity and allow the athlete to train with a higher exercise quality as indicated by kinetic and kinematic variables. It may be hypothesized, then, that these acute responses might be magnified or manifested in long-term performance

changes if these techniques are used in appropriately designed periodized training models.

LONG-TERM RESPONSES TO CLUSTER SET TRAINING

In one of the first studies on the topic, Byrd et al. (2) examined the effects of 10 weeks of resistance training with 3 different interrepetition rest intervals (zero, 1, and 2 seconds). In this study, it

was determined that the 2 groups that used rest periods between repetitions were able to increase their overall work output. While the major focus of this study was directed at cardiovascular adaptations, the results are important due to the relationship between total work and training adaptations. Frobose et al. (7) suggest that the adaptive response that stimulates muscular growth is more dependent on the

overall work output or volume load of the training session than the extent of the load, which appears to stimulate greater neural adaptations. Based on these findings, the implementation of the cluster set appears to allow the athlete to train with a higher intensity, thus magnifying the potential training stimulus affecting neural adaptations. Furthermore, this configuration allows fatigue aftereffects to diminish such that volume load can be increased, again, potentially magnifying adaptations.

Rooney et al. (18) also examined the effects of implementing different set configurations across a 6-week training program on isometric and dynamic markers of elbow flexor strength. Subjects were divided into 3 groups: (a) a control group that did no training, (b) a traditional set protocol, and (c) a cluster set protocol that used a 30-second interrepetition rest interval. Training was conducted 3 days per week with intensities and volumes varying between 6 sets of 6RM and 10 sets of 6RM loads. There were no differences between the cluster and traditional set protocols for maximal isometric strength. However, the traditional set configuration resulted in a significantly greater increase in dynamic muscular strength than the cluster set protocol. The results of this study suggested that during a 6-week elbow flexion protocol (18) the cluster set offered no benefit over the traditional paradigm. These results need to be examined carefully as they may be misleading in that power output was not measured. Lawton et al. (15) have suggested that cluster set training is best suited for explosive or ballistic exercises. Therefore, it is not unexpected that the elbow flexor exercise did not benefit from the cluster set protocol. Additionally, the subject population was relatively untrained, consisting of 18- to 35-year-old males and females who were only defined as being healthy. Plisk and Stone (17) in a recent review on periodization suggested the implementation of a cluster set paradigm is best suited for trained or highly trained individuals. Based on

these contentions, the results of the study by Rooney et al. (18) are not unexpected, and the findings may be different if highly trained athletes were used in conjunction with explosive exercises performed in a cluster fashion.

To the authors' knowledge, the only training study that examines the effects of varying the set structure with athletes was performed by Lawton et al. (14). In this study, 12 junior basketball and 14 junior soccer players with a minimum of 6 months of resistance training experience were divided into 3 training groups. Subjects were divided into 2 training groups: (a) a traditional set group in which 4 sets of 6 repetitions performed at a 6RM intensity every 260 seconds and (b) a cluster set group of 8×3 performed at a 6RM intensity every 113 seconds. Subjects trained the bench press 3 days per week for 6 weeks with ~24 total repetitions for ~13 minutes and 20 seconds of exercise. At the completion of the study, the traditional set training intervention resulted in significantly greater power outputs and 6 RM strength when compared to the cluster set group. Additionally, the traditional set group resulted in a significantly greater time under tension, which was hypothesized as one of the key reasons why the traditional set produced superior results. However, the results reported by Lawton et al. (14) may have occurred as a result of the training program intervention used by the traditional group being identical to the testing protocol used to evaluate changes in muscular strength as indicated by a 6RM. This contention is supported by data presented by Izquierdo et al. (11) that suggest that when training is conducted using sets to failure, a greater improvement in tests of muscular endurance (repetitions to failure) is noted. Since Lawton et al. (14) used a 6RM, which might be considered a high-intensity muscular endurance test (25), to evaluate performance, it would be expected that the traditional set configuration would result in superior performance gains after 6 months of practicing the performance

of a 6RM protocol. If other strength measures were used to assess performance gains, it is likely that different results would have occurred.

Collectively, when considering the body of knowledge about the use of cluster sets during long-term training, it appears that cluster sets allow the athlete to achieve higher power outputs and higher volume loads or work outputs. There is a paucity of long-term training intervention data looking at the effects of using the cluster paradigm with explosive or ballistic exercises. However, while the few studies currently available suggest the cluster set offers no long-term strength gain benefit, the strength and conditioning professional should consider the cluster set as a tool that may be useful in the development of specific athletic traits (9,14). This tool is probably best suited for ballistic explosive exercises and less useful for non-ballistic exercises such as the bench press. Additionally, the cluster set has yet to be investigated in the context of a periodized training program. Further research is needed in order to define the most effective time points during the periodized training program in which the cluster set is most beneficial.

PRACTICAL APPLICATION OF CLUSTER SET

The implementation of cluster sets in a periodized program can be accomplished in many ways depending on the specific goals of the phase of training. For example, the goals of the hypertrophy phase of training are to stimulate hypertrophy, decrease body fat mass, and increase work capacity (23). The traditional set may actually be best suited for this phase of training for most exercises, but the goals of the hypertrophy phase of training may also be accomplished by employing shorter rest intervals, such as 15 seconds, in the cluster set. The duration of recovery between each repetition may depend on the complexity of the exercise in which the cluster set is being employed. Based on the contemporary literature, it appears that power exercises such as the power clean or

Table 3
Example cluster set implementation during a hypertrophy phase of training

Exercise	Sets × repetitions	Set type	Intensity		Interrepetition rest interval (s)
			kg	% 1RM	
Power snatch	1 × 10/1	Cluster	90	75	15
Snatch grip shoulder shrugs	3 × 10	Traditional	140	115*	0
Rest 15 min					
Power clean	1 × 10/1	Cluster	115	80	15
Clean pull (FL)	3 × 10	Traditional	160	110†	0
Clean grip RDL	3 × 10	Traditional	120	86†	0
Max power snatch = 120 kg; max power clean = 145 kg; RDL = Romanian deadlift. *Based on maximum power snatch. †Based on maximum power clean.					

power snatch might be most affected by using the cluster paradigm. For example, in weightlifting, it has been argued that performing traditional sets with the complete lifts (i.e., clean, snatches, power cleans, power snatches) using high-repetition schemes results in fatigue-induced alterations in technique that may result in the development of technical deficiencies (9,12). As a result of this belief, weightlifters generally only perform repetition schemes, which range between 1 and 5 repetitions per set (12). Conversely, the cluster set may

offer a desirable solution to the volume limits that are sometimes placed on performing complete lifts as it results in increased work tolerance and can help maintain or enhance performance outcomes. Table 3 gives an example of a strength-endurance phase of training in which cluster sets could be employed for the power snatch and power clean, while traditional sets are performed with less technical exercises such as clean pulls. While the example presented in Table 3 uses 10 singles performed as a cluster of singles, one

might also consider performing clusters of 2 repetitions for a total of 10 repetitions for the given set. This type of cluster orientation may actually result in greater increases in endurance as there is less recovery between each repetition.

When formulating a basic strength phase of a periodized training program, one of the primary goals is to increase muscular strength; thus, using a traditional set configuration for nonballistic exercises may be warranted. However, among advanced athletes, the force/power production during ballistic movements can be augmented using cluster sets. A shift in the interrepetition rest interval to 30 seconds is warranted as the overall intensity of the cluster should be higher in this phase of training and a greater amount of recovery may be needed between each repetition of the set. In this phase of training, the strength and conditioning professional should consider implementing an undulating cluster set configuration as it will allow the athlete to begin to use higher lifting intensities. When using the undulating cluster set, the overall intensity of the set is defined as the average kilograms lifted across the set. For example, if the athlete is to perform an undulating cluster set of 5 repetitions with an average intensity of 110 kg, the athlete might perform lifts at 105, 112.5, 117.5, 112.5, and 105 kg for each set. Table 4 presents an example of a training session in which the undulating cluster set it is used.

During a strength/power phase of a periodized training program, the primary goals are to maintain or increase muscular strength and improve power-generating capacity (23). This phase may be the logical time during a periodized program to fully use a cluster design. In this scenario, for example, the strength and conditioning coach could implement the undulating cluster set as this set configuration has a reasonable potential to produce postactivation potentiation effects. Furthermore, the use of a heavy load (mid-cluster) helps to maintain or stimulate strength gains. The use of

Table 4
Example cluster set implementation during a basic strength phase of training

Exercise	Sets × repetitions	Set type	Intensity		Interrepetition rest interval (s)
			kg	% 1RM	
Snatch grip shrugs	3 × 5	Traditional	160	133*	0
Power snatch	3 × 5/1	Undulating cluster (105, 112.5, 117.5, 112.5, 105)	110	92*	30
Snatch pull	3 × 5	Traditional	145	120*	0
Snatch grip RDL	3 × 5	Traditional	125	104*	0
Max power snatch = 120 kg; max power clean = 145 kg; RDL = Romanian deadlift. *Based on maximum power snatch.					

Table 5
Example cluster set implementation during a strength/power phase of training

Exercise	Sets × repetitions	Set type	Intensity (kg)	Interrepetition rest interval (s)
Speed squats	3 × 3	Traditional	90	0
Power clean	1 × 3/1	Ascending cluster	110, 115, 120	30
	1 × 3/1	Ascending cluster	115, 120, 125	30
	1 × 3/1	Ascending cluster	120, 125, 130	30
Push jerk	3 × 3	Traditional	120	0

Based on max back squat = 180 kg; max power clean = 140 kg; max push jerk = 135 kg.

a 30- to 45-second rest interval in conjunction with either the undulating (or ascending cluster) may be useful in allowing the athlete enough recovery between repetitions so that higher power outputs are achieved. In the ascending cluster set, the athlete would progressively increase the overall intensity of the set with each set. Thus, force output could be maximized on the final repetition. Several different variations of ascending clusters could be used. For example, the athlete may perform 3 sets of clusters consisting of 3 repetitions with the average intensity of each set being increased (set 1 = 110, 115, 120; set 2 = 115, 120, 125; set 3 = 120, 125, 130). This method potentially emphasizes peak force development and may be more suited to partial movements or power movements (e.g., power snatch) rather than full weightlifting movements due to a fatigue effect using 3 ascending sets in a row. Other configurations could be constructed potentially emphasizing different characteristics. Table 5 presents an example of a training session using an ascending cluster set.

CONCLUSIONS

Based on theoretical and actual scientific data, the cluster set appears to be a unique method for introducing training variation into the periodized training program. The various methods for implementing cluster sets offer the strength and conditioning professional a tool that may be useful when working with training and highly trained athletes.

While more research needs to be conducted examining the performance and physiological affects of the various cluster set models, current data suggest that strength and conditioning professionals should consider using this novel training stimuli as part of their training plans, especially when working with explosive exercise such as the power clean, power snatch, and potentially pulling exercises (clean and snatch).■

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